非完整约束机器人 怎么处理 控制力向量 $F(q)$

direction. If the robot is a wheeled mobile robot subject to rolling constraints $A(q)\dot{q} = 0$, however, the calculated $F(q)$ must be projected to controls $F_{\text{proj}}(q)$ that move the robot tangentially to the constraints. For a kinematic robot employing the control law $\dot{q} = F_{\text{proj}}(q)$, a suitable projection is

假设 velocity space 三维 $(\dot{x}, \dot{y}, \dot{\theta})$

对第二种 $\dot{q}=F(q)$ →

投影

$$F_{\text{proj}}(q) = \Big(I - A^{\mathrm{T}}(q)\big(A(q)A^{\mathrm{T}}(q)\big)^{-1}A(q)\Big)F(q).$$

约束 $A\dot{q}=0$ 定义这个空间

一个切空间(线面/曲面)

对 $F(q)$ 向这个面做投影

For a dynamic robot employing the control law $u = F_{\text{proj}}(q) + B\dot{q}$, the projection was discussed in Section 8.7.

投影矩阵 $P = I - A^T(AA^T)^{-1}A$

10.6.5 Use of Potential Fields in Planners

把势场当作 A^* 启发式函数.

A potential field can be used in conjunction with a path planner. For example, a best-first search such as A^* can use the potential as an estimate of the cost-to-go. Incorporating a search function prevents the planner from getting permanently stuck in local minima.

最大程度保留原始力的方向信息

10.7 Nonlinear Optimization

所有 motion planning 问题 $\xrightarrow{\text{表述}}$ 非线性优化问题

The motion planning problem can be expressed as a general nonlinear optimization, with equality and inequality constraints, taking advantage of a number of software packages to solve such problems. Nonlinear optimization problems can be solved by gradient-based methods (最小梯度), such as sequential quadratic programming (SQP), or non-gradient methods (不用梯度), such as simulated annealing, Nelder–Mead optimization, and genetic programming. Like many nonlinear optimization problems, these methods are not generally guaranteed to find a feasible solution when one exists, let alone an optimal one. For methods that use gradients of the objective function and constraints, however, we can expect a locally optimal solution if we start the process with a guess that is "close" to a solution.

The general problem can be written as follows:

控制轨迹 状态轨迹

motion planning 标准优化格式

$$\begin{aligned}
&\text{find} && u(t), q(t), T && (10.6)\\
&\text{minimizing} && J(u(t), q(t), T) \quad \text{目标函数} && (10.7)\\
&\text{subject to} && \dot{x}(t) = f(x(t), u(t)), \quad \text{运动方程} \quad \forall t \in [0, T], && (10.8)\\
&\qquad && u(t) \in \mathcal{U}, \quad \text{(控制)} \quad \text{(动力学)} \quad \forall t \in [0, T], && (10.9)\\
&\qquad && q(t) \in \mathcal{C}_{\text{free}}, \quad \text{(避障)} \quad \forall t \in [0, T], && (10.10)\\
&\qquad && x(0) = x_{\text{start}}, \quad \text{(起点)} && (10.11)\\
&\qquad && x(T) = x_{\text{goal}}. \quad \text{(终点)} && (10.12)
\end{aligned}$$

正则约束

$$\begin{cases} J = T & \text{时间最优} \\ J = \int_0^T \|u(t)\|^2 dt & \text{能量最优} \\ J = \int_0^T \|\ddot{q}(t)\|^2 dt & \text{最平滑} \end{cases}$$

不要求轨迹上每点满足约束，只在有限个配点上检查约束

对 $u(t)$, $q(t)$ 和约束式进行离散化

To solve this problem approximately by nonlinear optimization, the control $u(t)$, trajectory $q(t)$, and equality and inequality constraints (10.8)–(10.12) must be discretized. This is typically done by ensuring that the constraints are satisfied at a fixed number of points distributed evenly over the interval $[0, T]$ and choosing a finite-parameter representation of the position and/or control histories. We have at least three choices of how to parametrize the position and controls:

参数化轨迹 (a) **Parametrize the trajectory $q(t)$.** In this case, we solve for the trajectory parameters directly. The controls $u(t)$ at any time are calculated using the equations of motion. This approach does not apply to systems with fewer controls than configuration variables, $m < n$.

参数化控制 (b) **Parametrize the control $u(t)$.** We solve for $u(t)$ directly. Calculating the state $x(t)$ requires integrating the equations of motion.

(Direct Collection) 直接配点法

同时参数化 $q(t)$ $u(t)$ (c) **Parametrize both $q(t)$ and $u(t)$.** We have a larger number of variables, since we parametrize both $q(t)$ and $u(t)$. Also, we have a larger number of constraints, as $q(t)$ and $u(t)$ must satisfy the dynamic equations $\dot{x} = f(x, u)$ explicitly, typically at a fixed number of points distributed evenly over the interval $[0, T]$. We must be careful to choose the parametrizations of $q(t)$ and $u(t)$ to be consistent with each other, so that the dynamic equations can be satisfied at these points.

A trajectory or control history can be parametrized in any number of ways. The parameters can be the coefficients of a polynomial in time, the coefficients of a truncated Fourier series, spline coefficients, wavelet coefficients, piecewise constant acceleration or force segments, etc. For example, the control $u_i(t)$ could be represented by $p + 1$ coefficients a_j of a polynomial in time:

$$u_i(t) = \sum_{j=0}^{p} a_j t^j.$$

In addition to the parameters for the state or control history, the total time T may be another control parameter. The choice of parametrization has implications for the efficiency of the calculation of $q(t)$ and $u(t)$ at a given time t. It also determines the sensitivity of the state and control to the parameters and whether each parameter affects the profiles at all times $[0, T]$ or just on a finite-time support base. These are important factors in the stability and efficiency of the numerical optimization.